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Design and Implementation of a Controlled Text Generation System for Product Descriptions

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Introduction

In recent years, the rapid growth of digital platforms and e-commerce systems has significantly increased the demand for high-quality product descriptions. Product descriptions play a crucial role in influencing customer decisions, improving user engagement, and enhancing the overall marketing strategy of businesses. However, writing product descriptions manually is a time-consuming task that requires consistency, creativity, and attention to detail. Traditional approaches to content creation rely heavily on human effort, which may lead to inconsistencies in writing style and quality. In addition, maintaining uniformity across multiple products can be challenging, especially when dealing with large datasets or product catalogs. Therefore, there is a need for automated systems that can generate accurate, structured, and controlled product descriptions. This project introduces a controlled text generation system designed to automate the creation of product descriptions. The system focuses on providing flexibility and control to the user by allowing them to define specific constraints such as required keywords, forbidden words, writing tone, and target audience. By doing so, the system ensures that the generated text meets the desired requirements while maintaining readability and coherence.

Project Idea

The main idea behind this project is to develop a system that can generate product descriptions automatically while maintaining strict control over the content. Unlike traditional text generation systems that produce random or uncontrolled outputs, this system is designed to follow user-defined constraints.

The system allows users to specify required words that must appear in the generated text, as well as forbidden words that should be avoided. Additionally, users can select the desired tone of the text, such as professional, marketing, luxury, or academic. This level of control enables the system to produce customized descriptions tailored to specific needs. The project is based on the concept of controlled generation, which ensures that the output is not only meaningful but also aligned with predefined rules. This approach simulates intelligent behavior without requiring complex or computationally expensive machine learning models.

System Design

The system is designed using a lightweight and efficient architecture that focuses on simplicity and functionality. It consists of three main components: input processing, text generation, and evaluation. The input processing stage collects user data, including product name, category, target audience, required words, forbidden words, and tone. This information is then used to guide the generation process.

The text generation component uses predefined templates and structured sentence patterns to create meaningful descriptions. These templates are dynamically filled with user-provided inputs, ensuring that the generated text remains relevant and coherent. The evaluation component analyzes the generated text to ensure that it meets the specified constraints. It checks whether the required words are included and verifies that forbidden words are not present. Based on this analysis, the system assigns a score to each generated description.

Implementation

The system is implemented using the Python programming language due to its simplicity and flexibility. The user interface is developed using Streamlit, which provides an interactive and user-friendly environment for input and output operations. The backend logic is responsible for generating text, processing constraints, and evaluating results. The system uses string manipulation techniques and rule-based logic to construct meaningful sentences. The implementation focuses on efficiency and clarity, ensuring that the system performs well even on low-resource machines. One of the key advantages of this implementation is that it does not rely on heavy machine learning models. This makes the system lightweight, fast, and easy to deploy.

Features

The system includes several advanced features that enhance its functionality and usability. One of the main features is the ability to generate multiple product descriptions at once. This allows users to compare different outputs and select the most suitable one. Another important feature is constraint control. Users can define required and forbidden words, ensuring that the generated content meets specific requirements. The system also supports multiple writing tones, providing flexibility in style and presentation. In addition, the system includes an evaluation mechanism that assigns a score to each generated description. This score reflects how well the text meets the defined constraints. The system also automatically selects the best result based on the highest score

Workflow

The system follows a structured workflow to ensure accurate and efficient text generation. The process begins with user input, where the user provides all necessary information about the product.

Next, the system generates multiple descriptions using predefined templates. These descriptions are then evaluated against the specified constraints. The evaluation process checks for the presence of required words and the absence of forbidden words. After evaluation, each description is assigned a score. The system then selects the best description based on the highest score and displays it to the user. Additional results are also presented for comparison.

Results

The system successfully generates clear and structured product descriptions that meet user-defined constraints. The generated text is readable, consistent, and suitable for real-world applications.

The evaluation system ensures that the output is reliable and accurate. In many cases, the generated descriptions achieve a score of 100%, indicating full compliance with the specified constraints.

The ability to generate multiple outputs allows users to explore different variations and choose the most appropriate one. This enhances the overall usability and effectiveness of the system.

Analysis

The analysis of the system shows that it is effective in generating controlled text while maintaining simplicity and efficiency. The rule-based approach ensures that the system remains lightweight and does not require significant computational resources.

The scoring mechanism plays a crucial role in evaluating the quality of the generated text. It provides a clear measure of how well each description meets the defined constraints.

Although the system does not use advanced AI models, it successfully simulates intelligent behavior through structured logic and rule-based processing.

Conclusion

This project demonstrates a practical and efficient approach to controlled text generation. By combining rule-based logic with user-defined constraints, the system is able to produce high-quality product descriptions.

The system improves productivity by reducing the time required for manual content creation. It also ensures consistency and accuracy, making it suitable for business and e-commerce applications.

Overall, the project highlights the importance of controlled generation in modern text processing systems.

Future Work

Future improvements to this system may include integrating advanced artificial intelligence models such as GPT or other deep learning techniques. This would enhance the naturalness and creativity of the generated text.

Additional features may include support for multiple languages, improved grammar correction, and more advanced constraint handling mechanisms. The system can also be expanded to support other types of content generation beyond product descriptions.